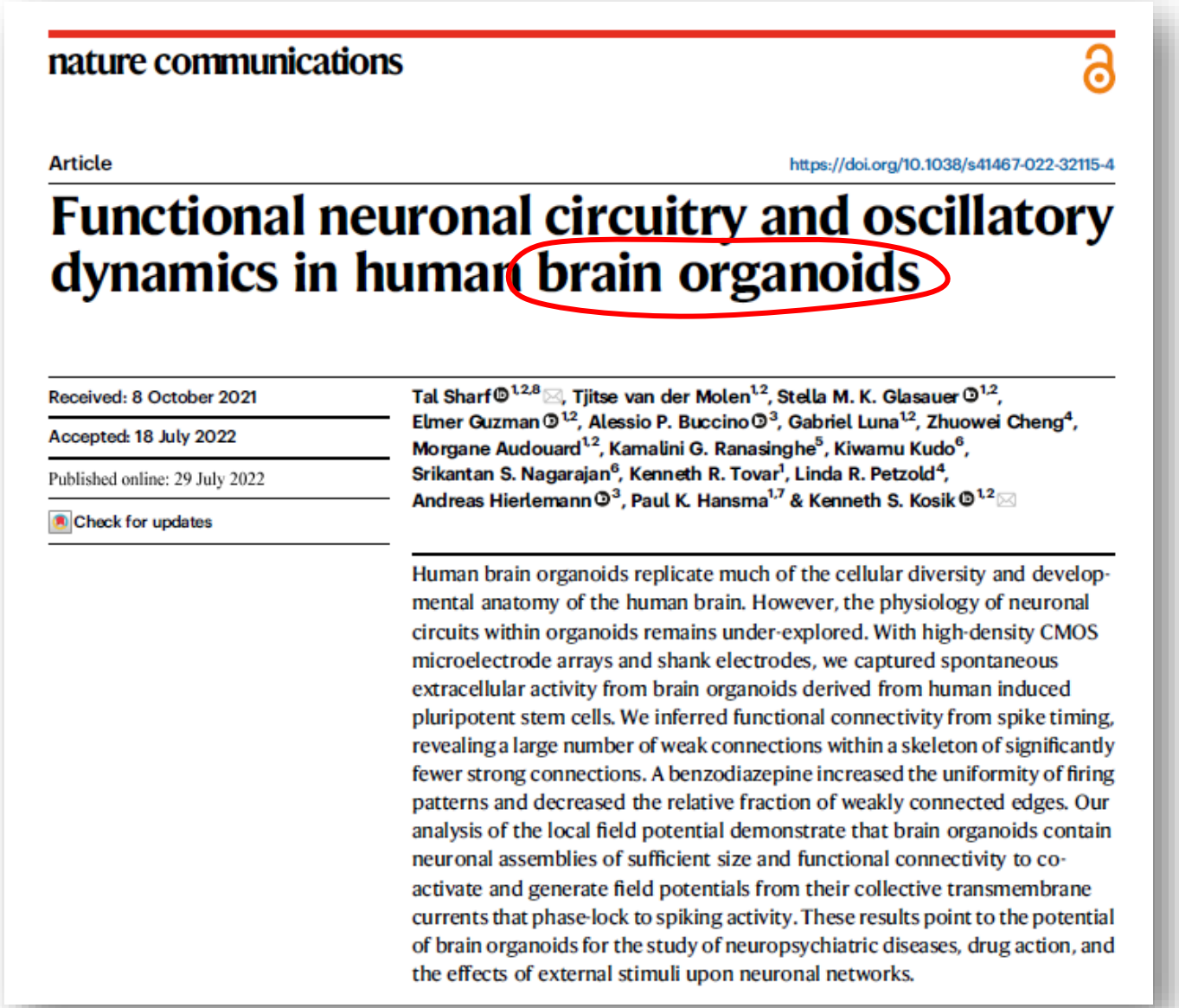
Single cell RNA-seq Analysis (scRNA)

Task:

Reproduce results from the publication: *Functional neuronal circuitry and oscillatory dynamics in human brain organoids*

Study in question:



nature communications

Article

https://doi.org/10.1038/s41467-022-32115-4

Functional neuronal circuitry and oscillatory dynamics in human brain organoids

Received: 8 October 2021

Accepted: 18 July 2022

Published online: 29 July 2022

Check for updates

Tal Sharf[1,2,8], Tjitse van der Molen[1,2], Stella M. K. Glasauer[1,2], Elmer Guzman[1,2], Alessio P. Buccino[3], Gabriel Luna[1,2], Zhuowei Cheng[4], Morgane Audouard[1,2], Kamalini G. Ranasinghe[5], Kiwamu Kudo[6], Srikantan S. Nagarajan[6], Kenneth R. Tovar[1], Linda R. Petzold[4], Andreas Hierlemann[3], Paul K. Hansma[1,7] & Kenneth S. Kosik[1,2]

Human brain organoids replicate much of the cellular diversity and developmental anatomy of the human brain. However, the physiology of neuronal circuits within organoids remains under-explored. With high-density CMOS microelectrode arrays and shank electrodes, we captured spontaneous extracellular activity from brain organoids derived from human induced pluripotent stem cells. We inferred functional connectivity from spike timing, revealing a large number of weak connections within a skeleton of significantly fewer strong connections. A benzodiazepine increased the uniformity of firing patterns and decreased the relative fraction of weakly connected edges. Our analysis of the local field potential demonstrate that brain organoids contain neuronal assemblies of sufficient size and functional connectivity to co-activate and generate field potentials from their collective transmembrane currents that phase-lock to spiking activity. These results point to the potential of brain organoids for the study of neuropsychiatric diseases, drug action, and the effects of external stimuli upon neuronal networks.

Sharf, T., van der Molen, T., Glasauer, S.M.K. *et al.* Functional neuronal circuitry and oscillatory dynamics in human brain organoids. *Nat Commun* **13**, 4403 (2022). https://doi.org/10.1038/s41467-022-32115-4

What are brain organoids?

- Miniaturized and simplified version of the brain **grown in a lab from human stem cells**.
- **3D** and **self-organizing** cellular structures.
- **Not** brains, **not** conscious.

Why is it interesting to study brain organoids?

If neuronal functionality of brain organoids is similar to brain, they could be used for the study of **neuropsychiatric diseases**, **drug action**, and **the effects of external stimuli** on the neuronal networks.

Overview of the study

- Authors recorded the **electrical signals** from the neurons inside the brain organoids and studied the **electrical activity** of these neurons.

- They then applied **diazepam** (a common sedative) and measured how it altered the activity and connectivity patterns across the organoid.

- Do the cell types present in these organoids confirm that the observed electrical activity is biologically plausible.
  - e.g., In the brain, certain cells are responsible for neuronal and oscillatory behavior. Do we find these cells in the organoids?

Sharf, T., van der Molen, T., Glasauer, S.M.K. *et al.* Functional neuronal circuitry and oscillatory dynamics in human brain organoids. *Nat Commun* **13**, 4403 (2022). https://doi.org/10.1038/s41467-022-32115-4

To get started:

- Download data from NCBI-Gene Expression Omnibus (GEO):
  - Accession number: GSE207749

- Data information:
  - Pre-processing already done
  - Count matrices available
  - Very sparse matrix

	Cell1	Cell2	Cell3	Cell4
Gene1	1	0	0	1
Gene2	1	50	0	111
Gene3	0	0	2	0

- Install required libraries:
  - Seurat: https://satijalab.org/seurat/articles/install_v5

Filtering steps

1. < 300 genes: Empty droplets, dead or dying cells, ambient RNA

2. > 10% mitochondrial gene content: Damaged cells, cytoplasmic RNA leaks out, but mitochondria (due to their own membrane) hold onto their RNA longer and these are sequenced.

Normalization

Why?

1. Remove the influence of sequencing depth
2. Make cells comparable to each other

How?

```
> NormalizeData(merged, normalization.method = "LogNormalize", scale.factor = 10000)
```

1. Cell gene count / total counts in that cell
2. Multiply by the scale factor
3. log(x+1)

$$Normalized_count = \log 1p(\,(count\ /\ total_counts) \times 10000\,)$$

Finding top variable features (genes)

Why?

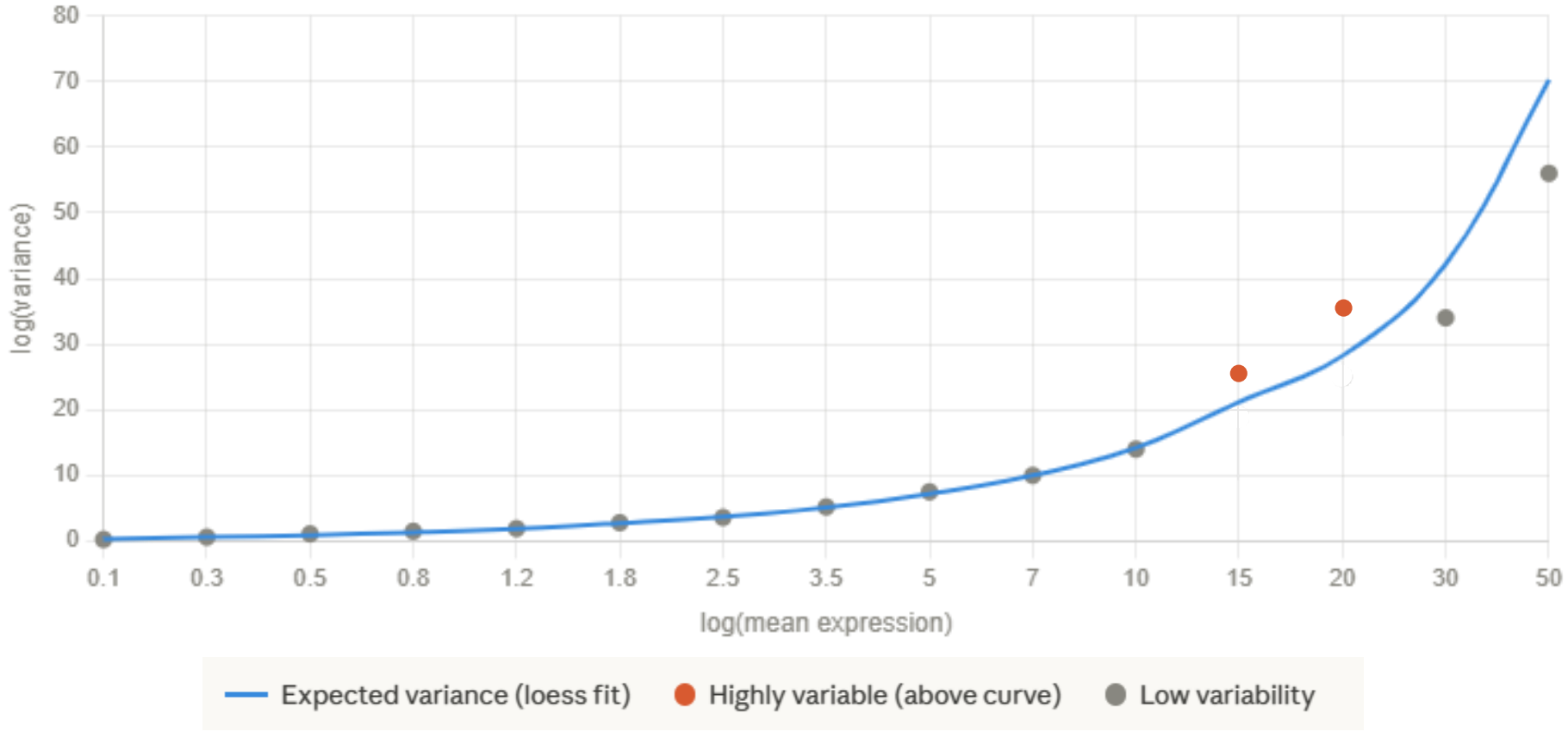
1. Performing downstream analysis on all the genes is **computationally very expensive**.
2. Most genes are expressed consistently across all cells or are expressed very lowly. They mainly add **noise** to the analysis.
3. We keep the genes that actually differ between cell populations.

How?

```
> FindVariableFeatures(merged, selection.method = "vst", nfeatures = 3000)
```

1. Calculates the **mean and variance** of each gene across all cells.
2. Fits a curve (**loess regression curve**) to model the expected variance given the mean (since *highly expressed genes naturally have higher variance*)
3. Finds and return top 3000 genes that are **more variable than expected** for their expression level.

Vst (Variance Stabilizing Transformation)



Scaling

Why?

1. Highly expressed genes dominate the downstream analysis (e.g. PCA).
2. Scaling converts expression values such that each gene has a mean of 0, and variance of 1 across cells.

How?

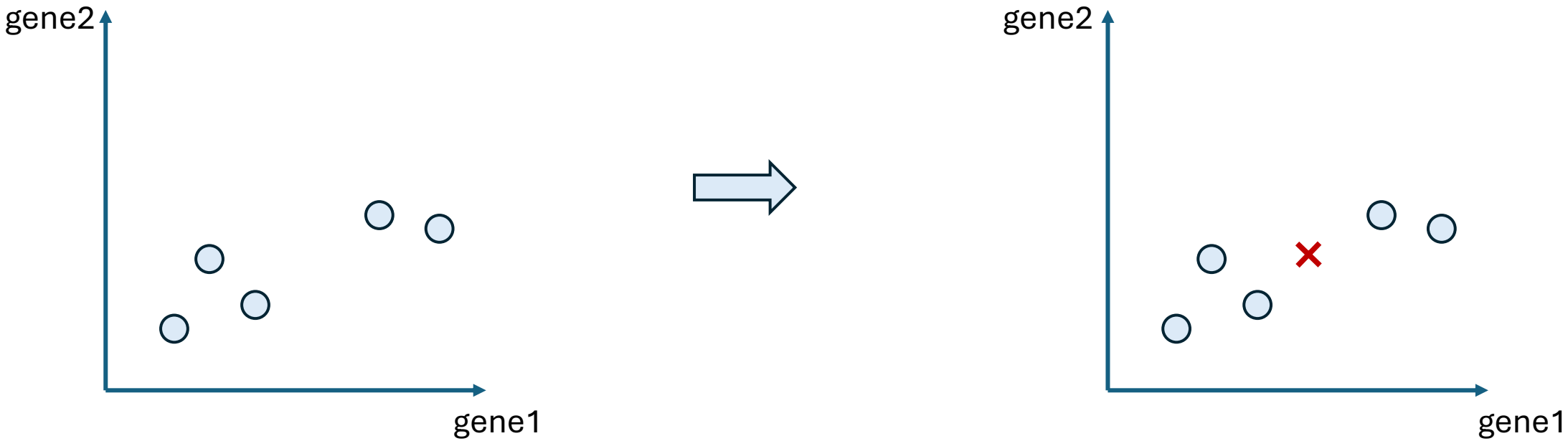
```
> ScaleData(merged)
```

- z-score normalization

$$Scaled_count = (value - mean) / std$$

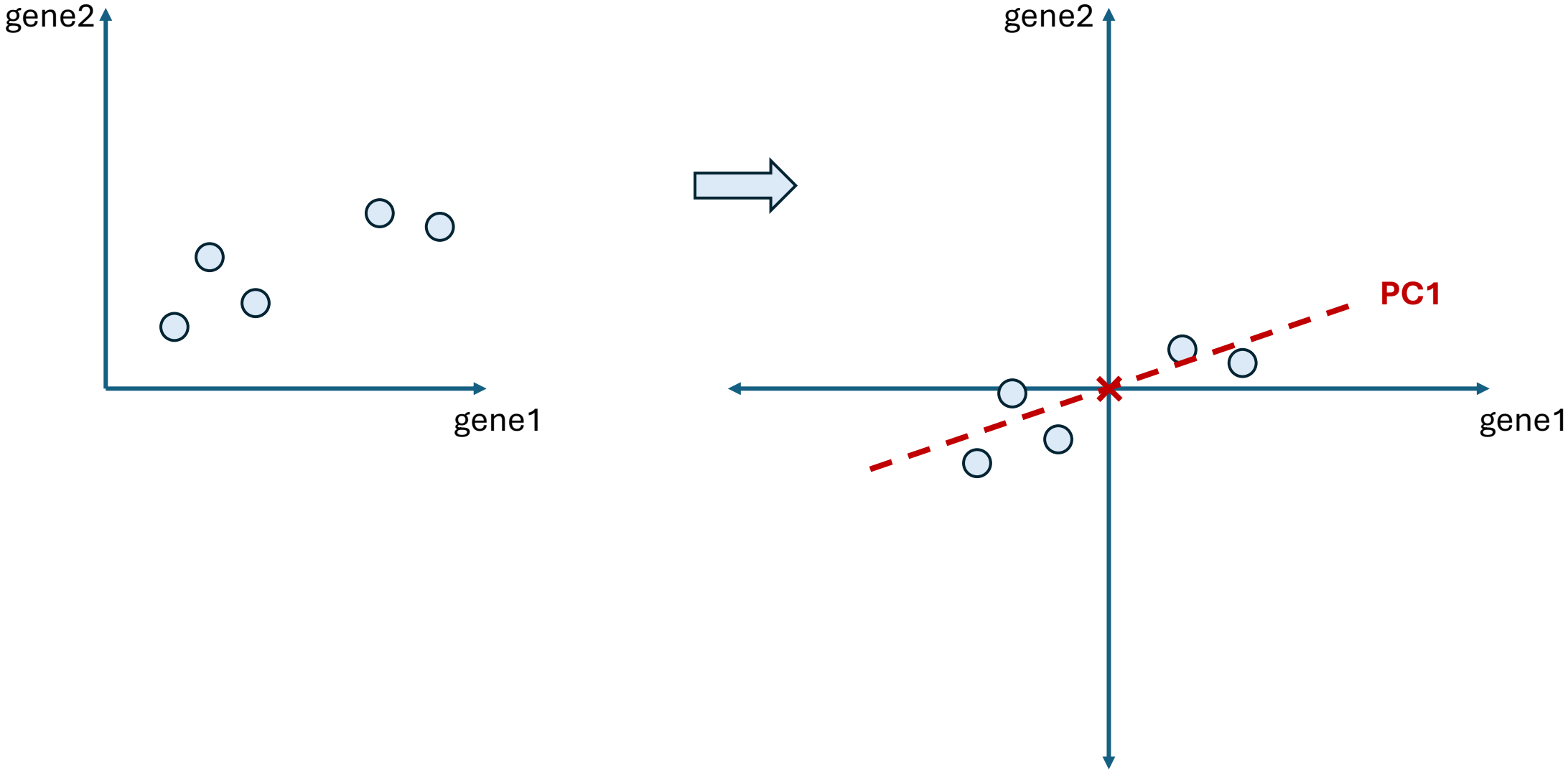
PCA (principal component analysis)

Linear combination of features (genes), that explain the variation in the data (droplets/cells).



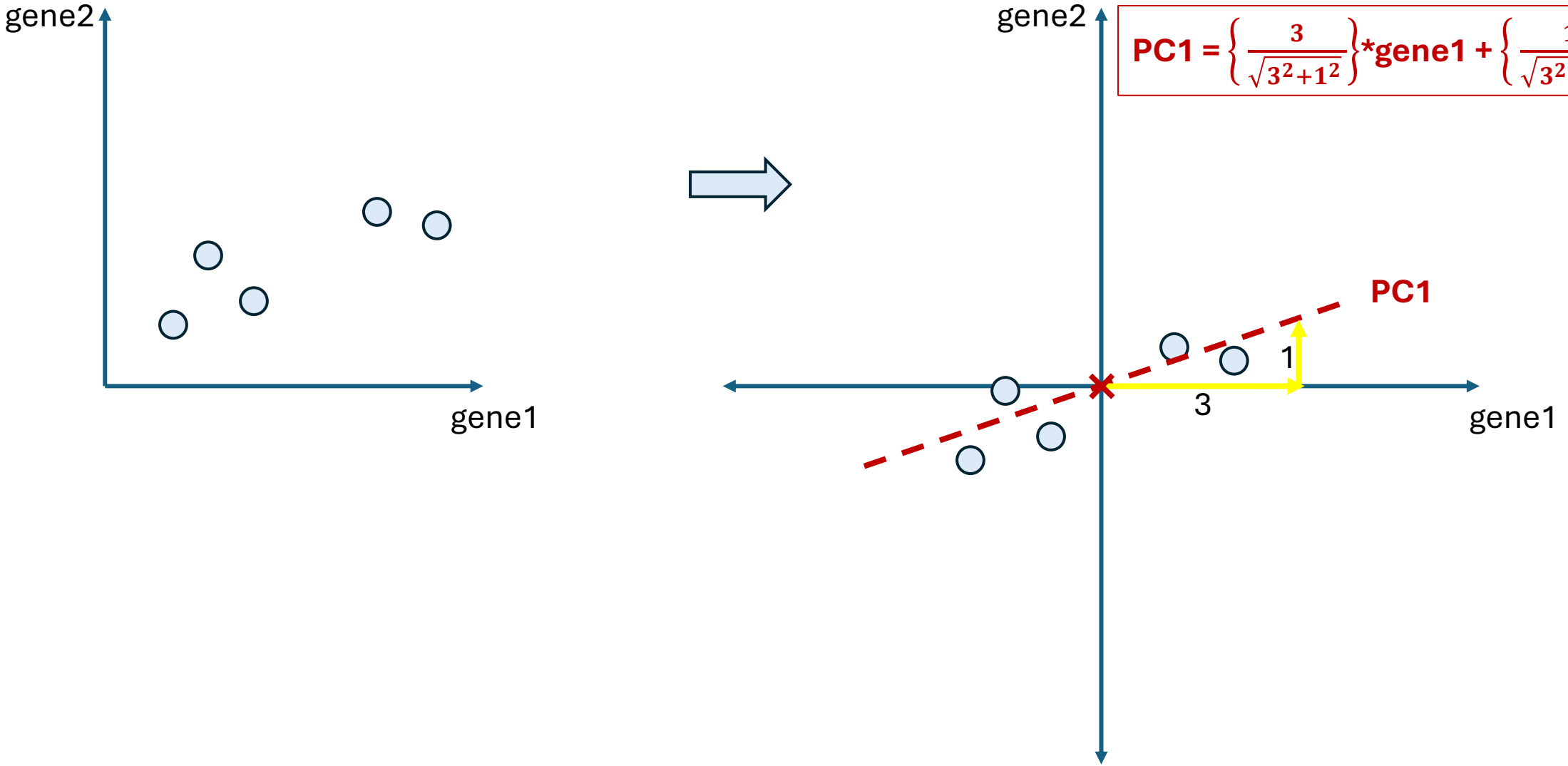
PCA (principal component analysis)

Linear combination of features (genes), that explain the variation in the data (cells).



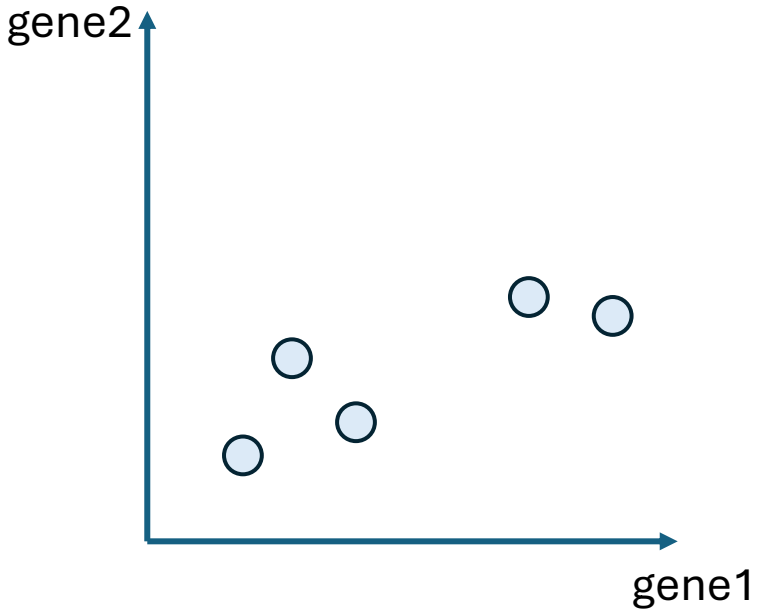
PCA (principal component analysis)

Linear combination of features (genes), that explain the variation in the data (cells).

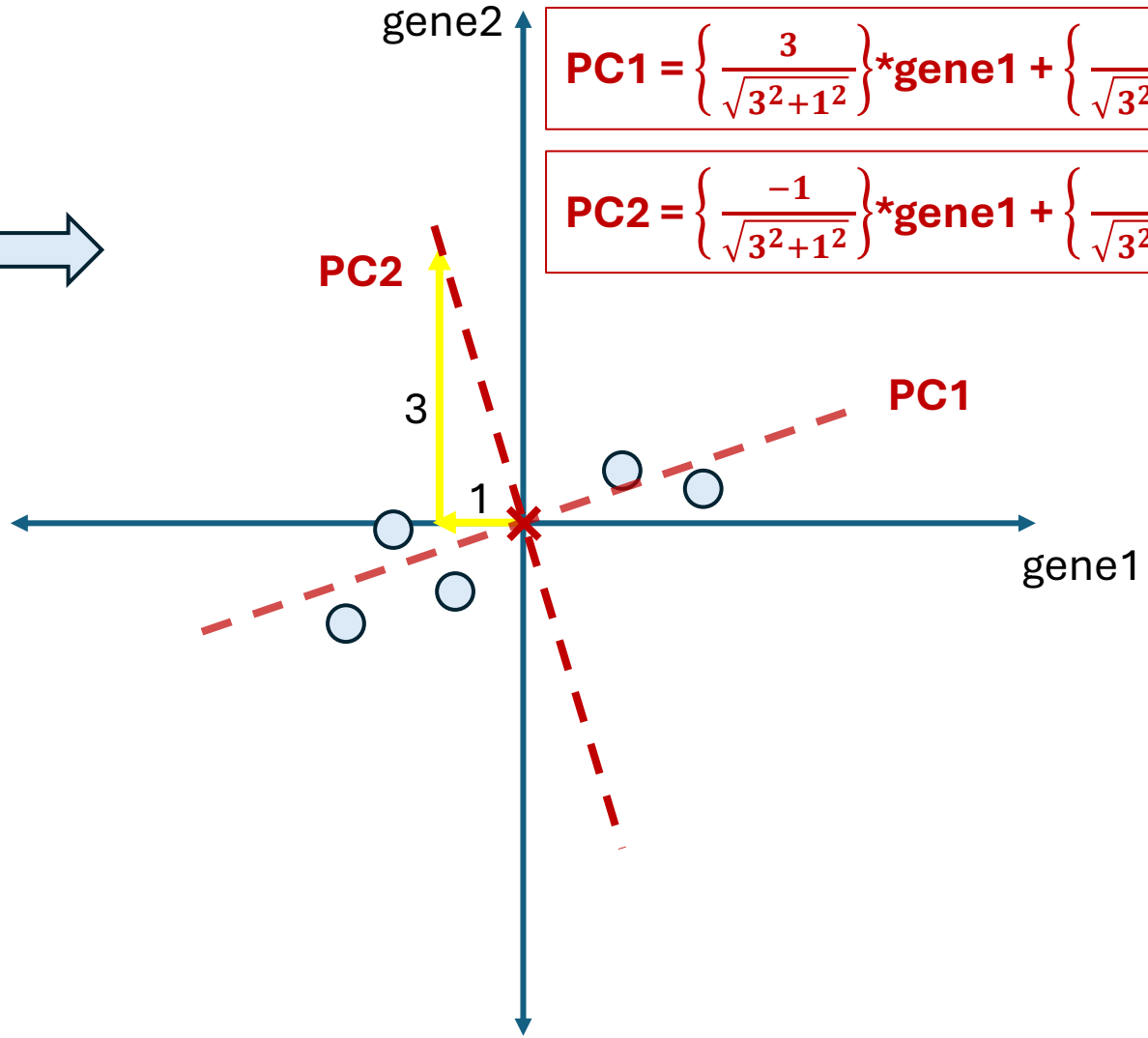


PCA (principal component analysis)

Linear combination of features (genes), that explain the variation in the data (cells).





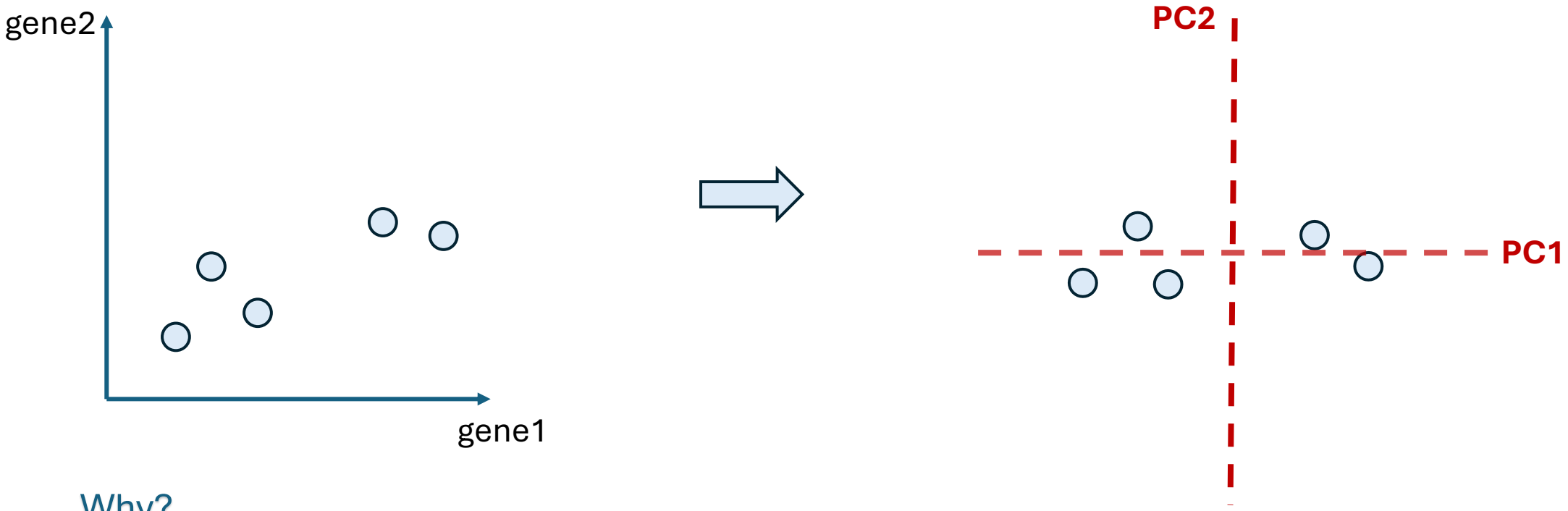


$$PC1 = \left\{\frac{3}{\sqrt{3^2+1^2}}\right\} * gene1 + \left\{\frac{1}{\sqrt{3^2+1^2}}\right\} * gene2$$

$$PC2 = \left\{\frac{-1}{\sqrt{3^2+1^2}}\right\} * gene1 + \left\{\frac{3}{\sqrt{3^2+1^2}}\right\} * gene2$$

PCA (principal component analysis)

Linear combination of features (genes), that explain the variation in the data (cells).



Why?

To reduce the dimensions to enable clustering of cells in the downstream analysis.

- Rather than 3000 genes, we'll consider top 30-40 PCs.
- Initial 10:50 PCs contain the most biological signals; rest of the PCs mainly induce noise.

How?

```
> merged <- RunPCA(merged, features = VariableFeatures(object = merged))
```

Clustering and Visualization

Why?

To find similar cell types.

How?

> FindNeighbors(merged, dims = 1:30)

- Find k (default k = 20) nearest neighbors.
- Create a KNN graph (k-nearest neighbors).
- Create an SNN graph (shared nearest neighbors).

> FindClusters(merged, resolution = 0.4)

- Uses Louvain algorithm to find clusters where cells inside are highly connected and cells outside are weakly connected.
- Resolution controls granularity.

  Low resolution : few, large clusters, high resolution: more, small clusters

> RunUMAP(merged, dims = 1:30)

What are doublets?

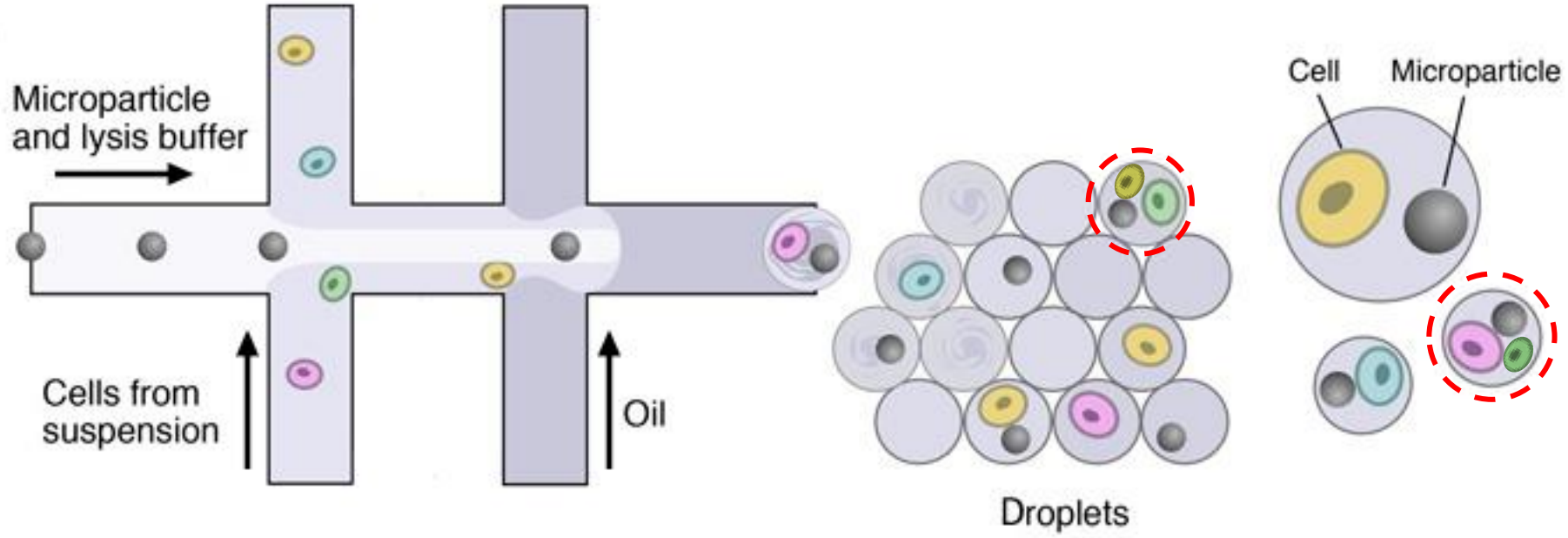


Image source: Macosko E, Basu A, Satija R et al **Highly Parallel Genome-wide Expression Profiling of Individual Cells Using Nanoliter Droplets** Cell, 161, 1202-1214

What are doublets?

Why?

- A doublet looks like a **mixture of two real cell** states and often sits between clusters in PCA space.
- It can distort clustering and cell-type inference.

How?

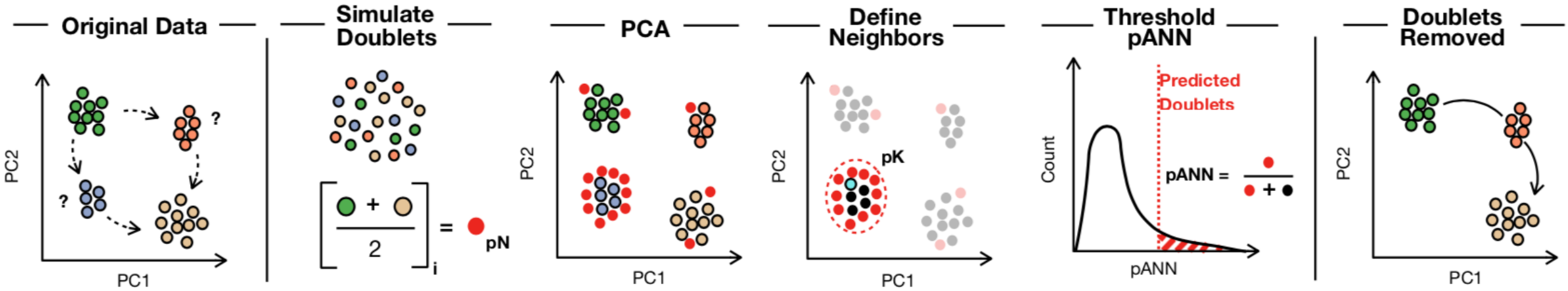
Using library "**DoubletFinder**"

(GitHub repo: https://github.com/chris-mcginnis-ucsf/DoubletFinder/tree/master)

```
> library(remotes)
> remotes::install_github('chris-mcginnis-ucsf/DoubletFinder', force = TRUE)
> library(DoubletFinder)
```

McGinnis C, Murrow L, Gartner Z **DoubletFinder: Doublet Detection in Single-Cell RNA Sequencing Data Using Artificial Nearest Neighbors** Cell Systems, 2019; 8, 329-337.e4

Finding and removing cell doublets (library: DoubletFinder)



For cell_i:

k = size of neighborhood

A = number of artificial doublets in the neighborhood

pANN_i (proportion of artificial nearest neighbors of cell_i) = **A / k**

Image source: https://github.com/chris-mcginnis-ucsf/DoubletFinder

Finding and removing cell doublets (library: DoubletFinder)

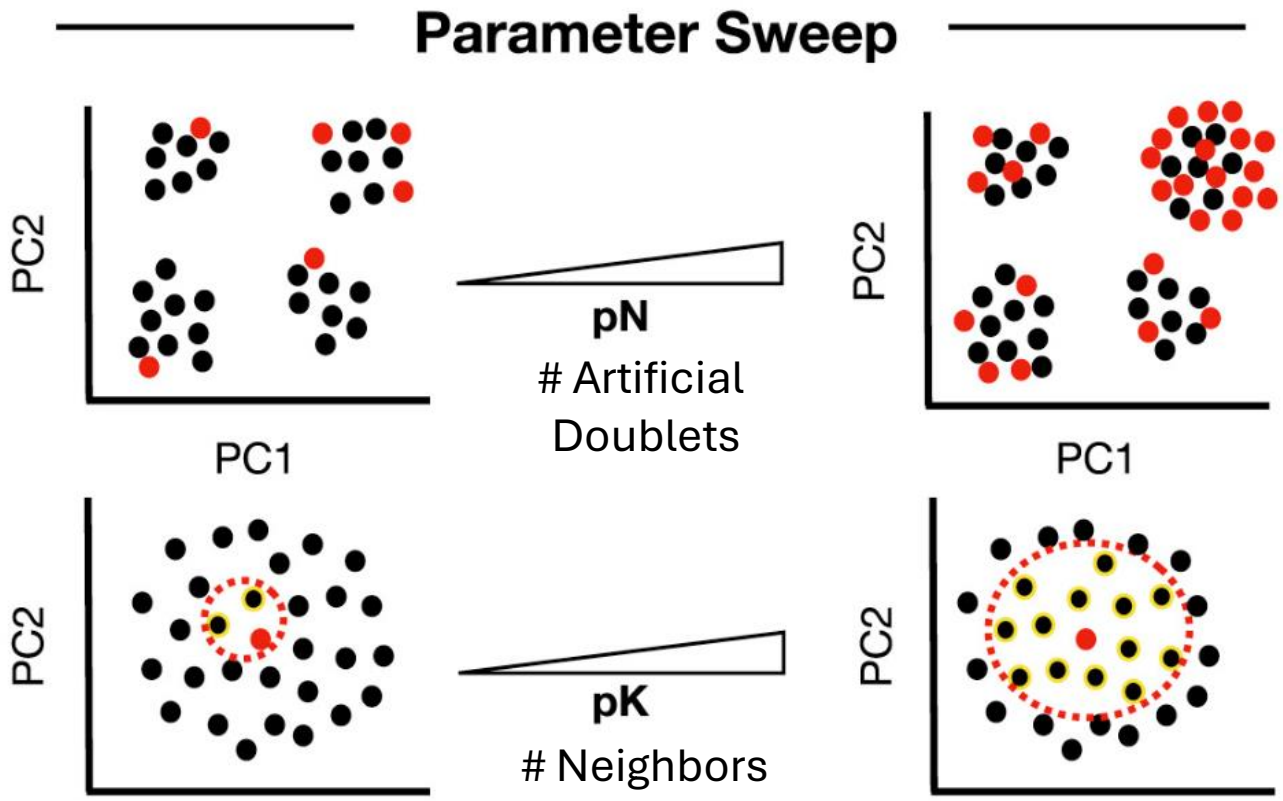


Image source: https://github.com/chris-mcginnis-ucsf/DoubletFinder

Finding and removing cell doublets (library: DoubletFinder)

Expected doublet rate: 5% (from literature)

Cells to be labelled doublets:

$$n_{doublet} = 0.05 * Number\ of\ total\ cells$$

Homotypic doublets correction: If two cells from the **same cell type** form a doublet, their average expression looks like a normal cell.

If $p_c = proportion\ of\ cells\ in\ cluster\ c$, then

$$homotypic.prop = \sum_c p_c^2$$

Adjust number of cells to be labelled doublets by removing the homotypic proportion:

$$\mathrm{n}_{\mathrm{doublet}}^{\mathrm{adj}} = n_{doublet} * (1 - homotypic.prop)$$

Example

Suppose clusters:

- A (50%)
- B (30%)
- C (20%)

Homotypic proportion:

$$0.5^2 + 0.3^2 + 0.2^2 = 0.38$$

Expected doublets:

$$0.05 \times 5000 = 250$$

Adjusted:

$$250 \times (1 - 0.38) = 155$$

We ask “doubletFinder” to find 155 doublets instead of 250.

Marker Genes

- Genes that are significantly differentially expressed in a cluster as compared to other clusters.

How?

```
> FindAllMarkers(seurat_new, test.use = "wilcox", logfc.threshold = 0.25, only.pos = TRUE)
```

Results

1. scRNA-seq analysis of three 7-month-old organoids identified 9 distinct cell populations, including ***excitatory neurons, GABAergic interneurons, astrocytes, and oligodendrocyte progenitors***.

2. Both excitatory and inhibitory neuron clusters expressed ***GABA***, ***AMPA***, and ***NMDA*** receptor genes, confirming the molecular machinery needed for network activity was present.

3. Expression of the forebrain marker ***FOXG1*** across all cell types, with minimal expression of retinal, midbrain, or hindbrain markers, established the organoids as a predominantly forebrain model.

4. The presence of parvalbumin-positive **GABAergic interneurons** provided molecular evidence that the organoids had the functional circuit architecture necessary to generate complex spiking patterns observed in the microelectrode array (MEA) recordings.

Links for further learning

- **Seurat library:**
  - **satijalab:** https://satijalab.org/seurat/

- **DoubletFinder:**
  - **Publication:** https://www.cell.com/cell-systems/fulltext/S2405-4712(19)30073-0
  - **GitHub repo:** chris-mcginnis-ucsf/DoubletFinder

- **Tutorials for scRNA:**
  - **YouTube:** @Collection_of_online_tutorials
  - **Seurat documentation:** https://satijalab.org/seurat/

- **Principal component analysis:**
  - **YouTube video**: StatQuest: Principal Component Analysis (PCA), Step-by-Step